



*The DNA of work*

# AI Job Market Impact Tracker

## User manual

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How the dashboard works, what every term means,  
and where every number comes from.  
Grounded in the Anthropic Observed Exposure methodology.  
Six regions - weekly refresh.

## Developed by AWA

Advanced Workplace Associates | Research instrument

# 1. What this tool is

The AI Job Market Impact Tracker is a research instrument that turns disparate signals about AI's effect on labour markets into a single, weekly-refreshed evidence base, viewable as a dashboard and downloadable as raw data. It covers six regions: the United States, United Kingdom, India, European Union, Asia Pacific, and Australia.

It exists because the public debate about AI and jobs lurches between two extremes — nothing is changing, or everything is. The reality, as Anthropic's own research shows, is that the early signals are subtle, narrow, and easy to miss in aggregate statistics. The tracker is built to detect those signals as they emerge.

## 1.1 Who it is for

Workplace strategy consultants, workforce planners at large employers, policy researchers, and journalists who need an accountable, sourced view of how the labour market is shifting.

## 1.2 What it is not

- It is not an oracle. It cannot predict which individual will lose a job.
- It is not a causal claim. It tracks correlations and attributes events to AI only where the employer or a credible source says so.
- It is not investment or hiring advice. It is a research tool.

# 2. Background research — what the methodology is built on

Massenkoff and McCrory (Anthropic, 5 March 2026) published a paper titled Labor market impacts of AI: A new measure and early evidence. It introduced a metric they called **Observed Exposure**, designed to measure not just which jobs are theoretically vulnerable to AI but which are already, in practice, being done by AI in real-world traffic.

## 2.1 The three inputs

Observed Exposure combines three independent data sources:

- **O\*NET**: the US Department of Labor's database of around 800 occupations, each decomposed into specific tasks.
- **Eloundou et al. (2023)**: an academic paper that scored every O\*NET task on a 0 / 0.5 / 1 scale (called  $\beta$ ) for its theoretical exposure to LLM speed-up.
- **The Anthropic Economic Index**: real-world usage data showing what tasks Claude is actually being asked to do, and whether the use is automated (AI does the work) or augmentative (human and AI collaborate).

## 2.2 The headline findings (US, as of March 2026)

- 97% of observed Claude task usage falls into categories Eloundou et al. rated as theoretically feasible ( $\beta \geq 0.5$ ).
- The most exposed US occupations are **Computer Programmers (75%)**, **Customer Service Representatives**, **Data Entry Keyers (67%)**, and **Financial Analysts**.

- 30% of US workers have zero exposure under this measure — their tasks appeared too infrequently in Claude usage to count.
- BLS 2024-2034 employment projections fall by 0.6 percentage points for every 10 pp of additional Observed Exposure.
- Top-quartile-exposure US workers are more likely to be female (+16 pp), more educated, older, and earn 47% more than the unexposed group.
- Aggregate unemployment in the top-quartile-exposure group has not increased significantly since late 2022, though the job-finding rate for workers aged 22 to 25 in exposed occupations has dropped about 14%.

Massenkoff and McCrory's central message is one of analytical modesty: AI's labour-market impact may be like the internet or trade with China rather than like the COVID shock. If so, it will not be obvious from headline unemployment numbers. A measure that tracks the narrowing gap between theoretical capability and observed deployment is more likely to detect the inflection early.

## 3. How Observed Exposure is calculated

The score is built bottom-up from tasks to occupations to occupation categories.

### 3.1 Task level

Every O\*NET task receives three flags:

- $\beta$  from Eloundou et al.: 0, 0.5, or 1.
- **Observed usage**: does this task appear with sufficient frequency in the Anthropic Economic Index, and is it work-related?
- **Use pattern**: of the observed usage, what share is automation versus augmentation?

A task's coverage equals 1 if  $\beta \geq 0.5$  and it has sufficient work-related Claude usage, weighted by 1.0 for automation and 0.5 for augmentation. Otherwise, coverage = 0.

### 3.2 Occupation level

An occupation's Observed Exposure is the time-weighted average of its task-level coverage scores. Time weights come from O\*NET's reported time-on-task data: tasks workers spend more of their day on count more.

### 3.3 Sector and regional aggregates

Occupation scores are aggregated to occupation categories (e.g., Computer & Math, Office & Admin) by current employment weight. For regions other than the US, the local occupation classification is crosswalked to O\*NET-SOC first; where local time-on-task data exists, it replaces the US weights.

**Robustness check.** The Anthropic paper varies several judgement calls (the  $\beta$  coding, the automation weight, the usage threshold) and reports that the Spearman rank correlation of occupation-level exposures across reasonable parameterisations is very high. The ranking is stable even when the absolute numbers move.

## 4. The KPI catalogue

The dashboard shows ten KPIs per region. Each KPI has a definition, a formula or source, a refresh cadence, and an alert band that triggers escalation in the weekly digest.

## AI-attributed layoffs (YTD)

|                   |                                                                                                                                                                                                              |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition</b> | Cumulative layoffs since 1 January where the employer cited AI. Measured for the US from the Challenger, Gray & Christmas monthly Job Cut Report; other regions remain modelled pending a comparable source. |
| <b>Source</b>     | Challenger, Gray & Christmas monthly report (US, measured); Layoffs.fyi tracker as context.                                                                                                                  |
| <b>Refresh</b>    | Monthly report, checked weekly.                                                                                                                                                                              |
| <b>Alert band</b> | More than 500 in a single week.                                                                                                                                                                              |

## Early-career unemployment delta (proxy)

|                   |                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition</b> | Youth minus overall unemployment rate, in percentage points, seasonally adjusted. A documented proxy for the top-quartile-exposure cohort: statistics agencies do not publish unemployment by AI exposure, so the original Massenkoff & McCrory diff-in-diff can only be replicated from CPS microdata. Measured: US 20-24 (BLS), UK 18-24 (ONS), EU under-25 (Eurostat), AU 15-24 (ABS). |
| <b>Source</b>     | BLS CPS, ONS LMS, Eurostat une_rt_m, ABS Labour Force; Massenkoff & McCrory 2026 framing.                                                                                                                                                                                                                                                                                                 |
| <b>Refresh</b>    | Monthly.                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Alert band</b> | More than 0.5 pp rise in the differential over 13 weeks.                                                                                                                                                                                                                                                                                                                                  |

## 22-25 hire rate

|                   |                                                                                                                           |
|-------------------|---------------------------------------------------------------------------------------------------------------------------|
| <b>Definition</b> | Year-on-year change in the monthly job-start rate among workers aged 22 to 25 in exposed occupations.                     |
| <b>Source</b>     | BLS CPS panel, ONS LFS, equivalent for each region. Brynjolfsson, Chandar & Chen (2025) provide the methodology template. |
| <b>Refresh</b>    | Monthly.                                                                                                                  |
| <b>Alert band</b> | More than 10% YoY drop.                                                                                                   |

## AI-mention posting share

|                   |                                                                                                               |
|-------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Definition</b> | Percentage of job postings whose description references AI, ML, GenAI, prompt engineering, or related skills. |
| <b>Source</b>     | Indeed Hiring Lab feeds, supplemented by Naukri JobSpeak (India) and Seek (Australia).                        |
| <b>Refresh</b>    | Weekly.                                                                                                       |
| <b>Alert band</b> | Doubling YoY.                                                                                                 |

## Capability gap

|                   |                                                                                                                                                                  |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition</b> | Theoretical $\beta$ (Eloundou) minus Observed Exposure, averaged across the region's occupations weighted by current employment. Expressed in percentage points. |
| <b>Source</b>     | Eloundou et al. 2023 plus Anthropic Economic Index.                                                                                                              |
| <b>Refresh</b>    | Monthly.                                                                                                                                                         |
| <b>Alert band</b> | Gap closes by more than 3 pp.                                                                                                                                    |

## Augmentation share

|                   |                                                                                                                                               |
|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition</b> | Percentage of Claude conversations in the region's traffic mix classified as collaborative (augmentative) rather than end-to-end (automated). |
| <b>Source</b>     | Anthropic Economic Index quarterly release.                                                                                                   |
| <b>Refresh</b>    | Quarterly.                                                                                                                                    |
| <b>Alert band</b> | Below 45% or above 60%.                                                                                                                       |

## Exposed-occupation posting index

|                   |                                                                                                            |
|-------------------|------------------------------------------------------------------------------------------------------------|
| <b>Definition</b> | Volume of job postings in the top-quartile-exposure occupations, indexed to a base of 100 in January 2025. |
| <b>Source</b>     | Indeed Hiring Lab plus Naukri JobSpeak and Seek; ONS Vacancy Survey for UK validation.                     |
| <b>Refresh</b>    | Weekly.                                                                                                    |
| <b>Alert band</b> | More than 5% drop on a 4-week rolling basis.                                                               |

## AI-skill salary premium

|                   |                                                                                                                                                  |
|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition</b> | Median advertised salary for postings mentioning AI minus median for comparable non-AI postings, expressed as a percentage of the non-AI median. |
| <b>Source</b>     | Indeed Hiring Lab plus Lightcast (formerly Burning Glass).                                                                                       |
| <b>Refresh</b>    | Monthly.                                                                                                                                         |
| <b>Alert band</b> | Premium widens or narrows by more than 10%.                                                                                                      |

## Graduate posting

|                   |                                                                                                                                                      |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition</b> | Year-on-year change in entry-level postings in exposed roles. Big 4 graduate disclosures (KPMG, Deloitte, EY, PwC) are incorporated where published. |
| <b>Source</b>     | Indeed Hiring Lab, IFOW graduate study, employer filings.                                                                                            |
| <b>Refresh</b>    | Quarterly.                                                                                                                                           |
| <b>Alert band</b> | More than 15% YoY drop.                                                                                                                              |

## Net AI-attributed job creation

|                   |                                                                                                                               |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Definition</b> | New AI, ML, data, security postings minus AI-attributed displacement events over the latest 12 months, in thousands of roles. |
| <b>Source</b>     | WEF Future of Jobs 2025 regional baseline plus posting-derived adjustments.                                                   |
| <b>Refresh</b>    | Quarterly.                                                                                                                    |
| <b>Alert band</b> | Net swings into negative territory.                                                                                           |

## 5. How to read each dashboard panel

### Most exposed occupations table

Ten occupations in the selected region with the highest Observed Exposure score. The blue bar is the score itself, 0 to 100%. The red bar is the capability gap — how much room the score has to grow before reaching theoretical  $\beta$  ceiling. When the red bar shrinks week on week without the blue bar staying flat, that is the earliest signal AI is moving deeper into that occupation.

### Net job change (created vs displaced)

A horizontal bar chart of jobs created (green) and displaced (red) over the last 12 months, by occupation family, in thousands of roles. Displacement figures filter to attribution-confidence of 3 or higher, so vague "AI-driven" press claims do not dominate. Use the source & method tab on the card to see which buckets the figures roll up from.

### Demographics in the exposed quartile

A doughnut chart of the age distribution of workers in the top-quartile-exposure occupations. The chart subtitle shows the female/male split. Compare across regions to see how exposure concentrates differently — for example, the Indian top-quartile skews far younger than the European one.

### Augmentation vs automation

A doughnut chart showing the share of Claude usage in the region that is collaborative versus end-to-end. A high augmentation share suggests workers are still in the loop; a high automation share suggests AI is doing more of the work alone. Anthropic's March 2026 release shows augmentation just over half of Claude.ai conversations.

### Posting volume in exposed roles

A 12-week indexed line showing how postings in the top-quartile-exposure occupations are moving. Base = 100 at the first week shown. A downward slope of more than 5% over a 4-week rolling window is the alert band.

### Capability gap by sector

Twin bars per sector: blue is theoretical  $\beta$ , red is Observed Exposure. The space between is headroom. This is the chart to watch over months and quarters — if red moves toward blue in a sector, that sector is converting capability into deployment.

### News & events feed

Most recent AI-attributed events in the region. Each is tagged with an attribution-confidence score: 5 = company filing or direct statement; 4 = first-party data such as Indeed or Naukri; 3 = credible reporting; 2 = secondary reporting; 1 = unsourced. Filter mentally to confidence  $\geq 3$  unless you are specifically tracking the narrative.

## 6. Regional adaptations

Observed Exposure was calibrated on US O\*NET tasks and US Claude traffic. Three adaptations let it work consistently across the six regions:

- **Crosswalking:** the local occupation classification is mapped to O\*NET-SOC. UK uses SOC2020 (ONS crosswalk), India uses NCO-2015 (Nasscom crosswalk), the EU uses ISCO-08 (CEDEFOP crosswalk), Australia and New Zealand use ANZSCO 2022.
- **Time-use reweighting:** where local time-on-task data exists, it replaces the US weights. For India, this is rare; for the EU, Eurostat LFS provides partial coverage.
- **Country brief substitution:** Anthropic has published country briefs for India and Australia and is rolling out more. Where a brief exists, its task-mix is used in place of global Claude data.

**Important caveat for cross-region reading.** Anthropic country briefs measure what Claude users in that country are doing – not what the country's workforce is doing. Cross-country comparisons should always use the within-country quartile rank as the primary comparable; absolute coverage is shown alongside as context only.

| Region         | Classification           | Statistical baseline                   | Real-time signal                                 |
|----------------|--------------------------|----------------------------------------|--------------------------------------------------|
| United States  | O*NET-SOC native         | BLS CPS, JOLTS, EP-2024-34             | Indeed Hiring Lab, Lightcast, layoffs.fyi        |
| United Kingdom | SOC2020 → O*NET          | ONS LFS, Vacancy Survey, IFOW          | Indeed Hiring Lab UK, LinkedIn Economic Graph UK |
| India          | NCO-2015 → O*NET         | NSO PLFS, Nasscom Strategic Review     | Naukri JobSpeak weekly, Foundit                  |
| European Union | ISCO-08 → O*NET          | Eurostat LFS, Cedefop Skills Forecast  | EURES, LinkedIn EU Economic Graph                |
| Asia Pacific   | Mixed (SSOC, JSCO, KSCO) | Singapore MOM, Japan MHLW, Korea KOSIS | JobStreet, Jobsdb, Wantedly                      |
| Australia      | ANZSCO 2022              | ABS Job Vacancies, Treasury            | Indeed Hiring Lab AU, Seek Employment Trends     |

## 7. Data sources and refresh cadence

Sources are tiered by reliability and latency. Lower tiers are used to colour and accelerate, never to overwrite higher-tier numbers.

| Tier                           | Sources                                                                                                    | Latency    | Role                                             |
|--------------------------------|------------------------------------------------------------------------------------------------------------|------------|--------------------------------------------------|
| Tier 1 — Methodological anchor | Anthropic Economic Index releases; Anthropic country briefs; Hugging Face dataset Anthropic/EconomicIndex. | Quarterly  | Sets Observed Exposure per occupation.           |
| Tier 2 — Official statistics   | US BLS, UK ONS, Eurostat, India NSO, Australia ABS, Singapore MOM.                                         | 1-3 months | Baseline employment, unemployment, demographics. |
| Tier 3 — Real-time market      | Indeed Hiring Lab, LinkedIn Economic Graph, Seek, Naukri JobSpeak, Lightcast.                              | Weekly     | Postings, AI-mention share, vacancy trends.      |

| Tier                       | Sources                                                                                      | Latency       | Role                                                        |
|----------------------------|----------------------------------------------------------------------------------------------|---------------|-------------------------------------------------------------|
| Tier 4 – Narrative & event | Layoffs.fyi, WEF Future of Jobs, McKinsey/IFOW/British Progress reports, central-bank notes. | Weekly-annual | AI-attributed layoffs, narrative, alternative attributions. |

## 7.1 Refresh cycle

The site auto-refreshes every Monday at 06:00 UTC via a GitHub Actions workflow. The workflow runs `scripts/update_data.py`, which pulls the latest values from each adapter, appends a new row per region per KPI to `data/historical.csv`, writes a frozen JSON snapshot to `data/snapshots/YYYY-Wxx.json`, and overwrites `data/current.json`. Netlify auto-deploys on the resulting commit.

## 7.2 Open data

Every value the dashboard has ever shown is downloadable as a single CSV (`historical.csv`) or as individual weekly JSON snapshots. Snapshots are never edited after the fact – if a source revises a figure, the new value enters a future snapshot, not the past one. This makes the data store an audit trail as much as a source.

# 8. Limitations and caveats

## 8.1 AI washing

Layoffs.fyi indicates roughly 48% of 2026 tech layoffs are "AI attributed" by the employer, while a stricter count puts the genuine-automation figure closer to 20%. The Hill and Crunchbase both quote analysts arguing that AI announcements are partly a cover for reallocating budgets to AI infrastructure, not for net headcount reduction caused by automation. Every AI-attributed layoff event in this tracker carries an attribution-confidence score (1 to 5) so reports can be filtered by stringency.

## 8.2 Survey lag and sampling

Anthropic Economic Index releases are quarterly and based on a one-week sample window. BLS, ONS, NSO, and Eurostat LFS data publish with a one to three month lag. The dashboard surfaces the freshness timestamp on every tile so users know what is live and what is structural.

## 8.3 Country-brief comparability

Anthropic country briefs measure usage by Claude users in that country, not the country's workforce. Cross-country absolute comparisons of Observed Exposure are misleading. The within-country quartile rank is the only safe comparable.

## 8.4 Scraping legality

LinkedIn's terms prohibit unauthorised scraping; only the licensable LinkedIn Economic Graph feed is used. Indeed Hiring Lab, Seek Employment Trends, Naukri JobSpeak, and equivalent aggregators provide ToS-compliant aggregated feeds. The pipeline never stores raw job postings; it stores aggregated metrics only.

## 8.5 Causal modesty

Even with the best current measures, the differential unemployment signal in the US is statistically indistinguishable from zero. This tool is built to detect movement as it emerges, not to assert causation prematurely. Every monthly and yearly report includes a section on alternative explanations — trade, policy, business cycle, demographic shift — that could account for the same observation.

## 9. References

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